

CLASH OF THE **CLAWS**

RV College of Engineering Team Pusle

**Pulse : An Autonomous Multi-Agent Shield Preventing
Human Error through Sandbox Verified AI
Orchestration**

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Problem Statement

Why we selected this theme (Theme 3: Productivity Platforms)

- **On-Device Sovereignty:** Uses local Ollama (Llama-3/CodeLlama) to keep sensitive infrastructure metadata and proprietary configs entirely offline.
- **Automated Velocity:** Employs an OpenClaw-style ReAct loop to automate security verification, eliminating manual DevOps approval bottlenecks.
- **GenAI Orchestration:** Researches multi-model collaboration (GPT-4o for speed, local Llama for depth) via Railway-Oriented Pipelines for high-stakes operations.

What problem are we solving?

- **Intent-Blind Execution:** Prevents "Fat-Finger" errors where simple typos in traditional shells lead to catastrophic production deletions.
- **Passive Logging Failure:** Moves beyond post-outage logging by using predictive intelligence to block commands based on their projected "Blast Radius."
- **The Agentic Risk Gap:** Provides a deterministic "Safety Brake" for autonomous agents to prevent accidental environment corruption during automated fixes.

Who faces it (The Personas)

- **Agile MSME Developers:** Small teams lacking enterprise-grade "four-eyes" approval systems for critical infrastructure.
- **Non-Technical Stakeholders:** Startup owners whose digital livelihood is vulnerable to a single misplaced command on a lone server.
- **Autonomous AI Systems:** Emerging DevOps platforms requiring a sandboxed "Ghost State" to verify generated scripts before deployment.

Why is it important?

- **Prevention Over Recovery:** Intercepts destructive intent at the terminal level before it reaches the kernel, treating the cause rather than the symptom.
- **Economic Continuity:** Prevents bankruptcy-level infrastructure wipeouts for small enterprises through predictive simulation.
- **Trust in Automation:** Provides a verifiable, agentic shield to ensure autonomous infrastructure remains controllable by humans.

Current Solution & Gaps

Existing Solutions (The Status Quo)

- **Static Permissions (IAM/RBAC):**
Binary systems like AWS IAM grant access based on identity ("Who") but fail to analyze the actual action ("What").
- **Reactive Monitoring(SIEM/Logs):**
Tools like Datadog log destruction after it happens alerting you to the fire but unable to stop the match.
- **Post-Disaster Recovery:**
Backup software treats the symptom of data loss, often resulting in significant downtime and data "drift" during restoration.

What's Missing?

- **Lack of Intent Analysis:**
Tools cannot distinguish between a safe `rm -rf tmp/` and a fatal `rm -rf /` because they don't understand a command's "Blast Radius."
- **The Permission Paradox:**
Developers need high-level access to be productive, but this creates a single point of failure where one typo can bankrupt a company.
- **Framework Bloat:**
Many DevOps AI agents are "black boxes" that prioritize completing tasks over the safety and integrity of the underlying system.

The Pulse Opportunity

- **Predictive Action Simulation:**
Introduces a "Ghost Sandbox" layer to validate intent by executing commands in isolation before they touch the production host.
- **Agentic Safety Guardrails:** Uses LLMs as a "Virtual Senior Engineer" to provide a second pair of eyes that understands the context of every operation.
- **Zero-Trust for the AI Era:**
Provides the essential "Verification Layer" needed to make AI-driven productivity safe for critical production environments.

Our Solutions

What we built: The Pulse Platform

- **Native Multi Model ReAct**

Orchestrator: A custom Python engine managing agentic reasoning via direct OpenAI and Ollama APIs, bypassing restrictive frameworks for maximum control.

- **Go Railway Execution Engine:** A high performance Go backend that handles shell AST parsing and sandbox orchestration to provide a deterministic safety layer for CLI operations.

- **Distributed Security Controller:** A cross language system bridging high level AI reasoning with low level kernel and filesystem security to create unified infrastructure guardrails.

Core Ideas:

- **Railway Oriented Functional Pipeline:**

A Go based functional pattern where command execution flows through "pure" agents, treating errors as first-class data on a separate track.

- **Stateless Session Management:**

Uses custom routing logic where agents maintain isolated histories and configurations, ensuring precise, task-specific cognitive boundaries.

- **Immutable State Snapshots:**

Binds every security decision to a unique, immutable context of the environment, creating a technically verifiable and unalterable audit trail.

Key Features:

- **Docker Alpine Sandbox:**

Leverages the Docker Engine to spin up ephemeral Alpine containers as a "Ghost State" for safe execution and observation of destructive commands.

- **Multi-Model Routing Logic:**

Intelligently dispatches orchestration to GPT-4o-mini while offloading deep risk assessments to local Llama-3 and CodeLlama via the Ollama API.

- **Shell AST Analysis:** Utilizes the `mvdan.cc/sh` parser in Go to perform Abstract Syntax Tree analysis, detecting malicious patterns and injections before they reach the sandbox.

Product Walkthrough

USER INTERACTION

Unified Command Entry (The Proposing Phase)

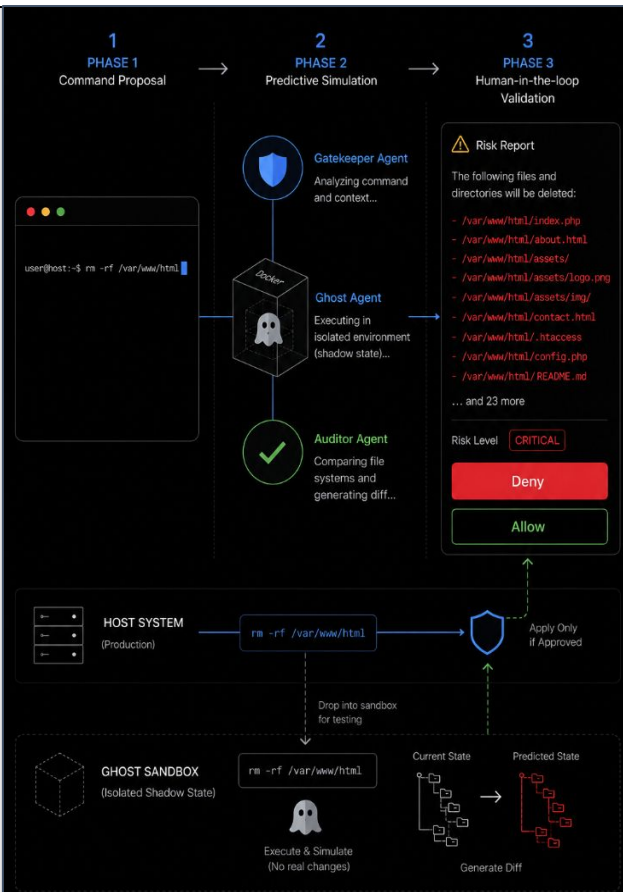
- **Contextual Terminal Input:** Commands are wrapped in immutable session contexts instead of executing immediately.
- **Implicit Intent Capture:** Captures command, directory state, and environment variables for agentic analysis.
- **High-Velocity Fallback:** Provides "Fast-Path" execution for low-risk commands, ensuring security never slows down productivity.

The Visual Pulse Audit (The Observation Phase)

- **Agentic Pipeline Status:** Offers real-time feedback through Gatekeeper, Ghost, and Auditor agents for a transparent experience.
- **Ghost-State Visualization:** Flagged commands trigger a "Simulation Preview," showing filesystem changes before they hit the host.
- **Deep Technical Advisory:** Advisor Agent (Llama-3) provides risk breakdowns and recommended self-healing steps when needed.

Decision Resolution (The Commitment Phase)

- **Interventionist Prompts:** Halts any command affecting critical paths, requiring a conscious "Allow" or "Deny" from the operator.
- **Audit Log Integration:** Finalizes interactions with a link to an immutable SQLite log for transparent review of interventions.
- **Actionable Self-Healing:** Denied actions include safe, alternative command suggestions to achieve goals without risk.



Product Walkthrough

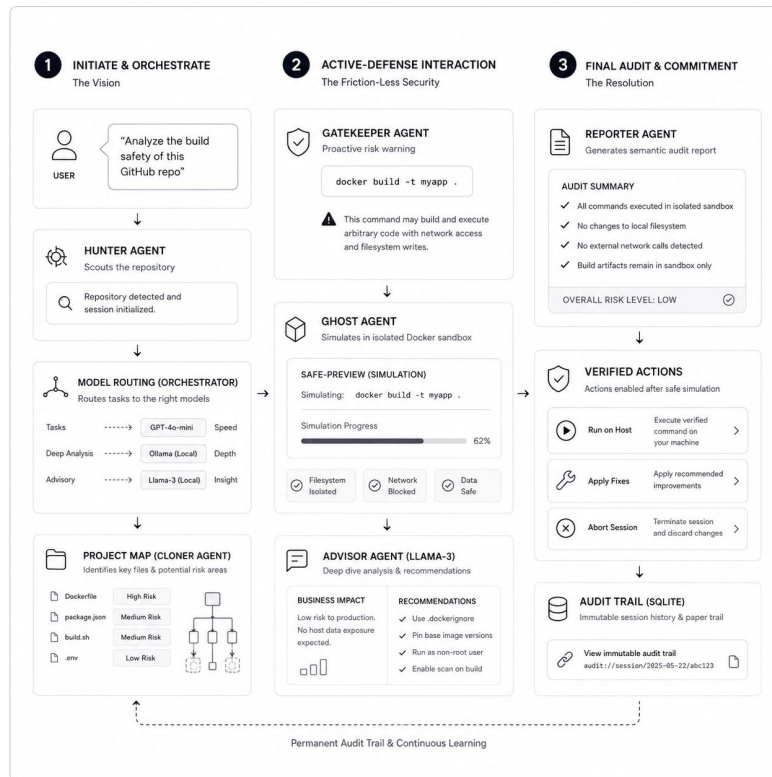
Key Moments: The Pulse Workflow

Intent Interception: The Gatekeeper Agent uses shell AST parsing to detect destructive patterns (e.g., recursive deletion, credential exposure) upon command submission. It instantly determines if the command is safe for the host or requires simulation, preventing human error from reaching the live kernel.

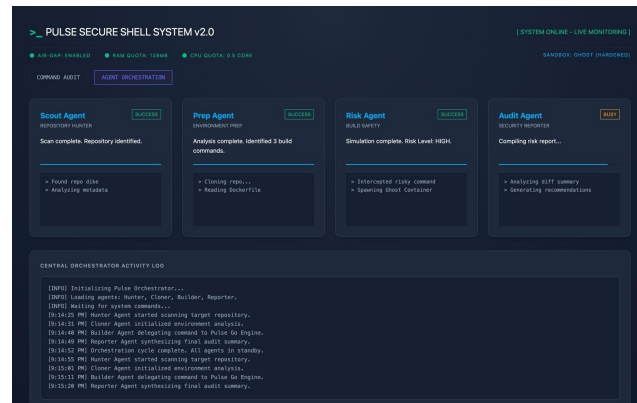
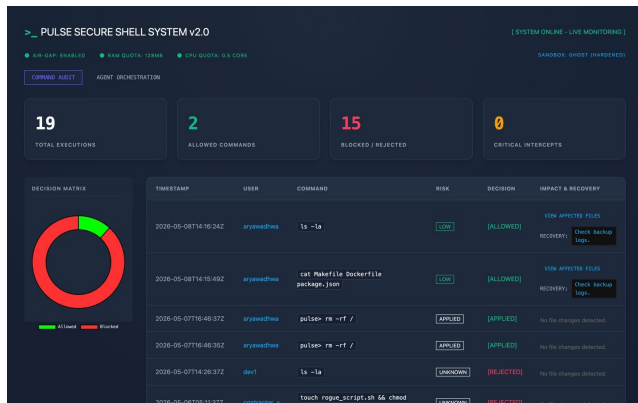
Ghost Simulation & Comparison: The Ghost Agent runs the command in an isolated Docker Alpine sandbox. The Auditor Agent performs a bitwise comparison between pre- and post-execution states to create an Impact Report.

Risk Revelation & Human Override: Pulse shows an interactive dashboard with a visual diff and risk score, enabling an evidence-based decision to commit or abort. The process is concluded with an immutable record in the SQLite Audit DB, documenting the reasoning and result.

User Experience Flow



Product Walkthrough



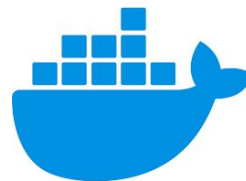
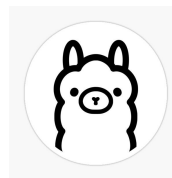
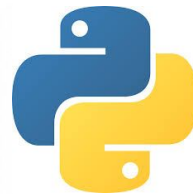
PULSE

[SECURE SHELL v2.0]

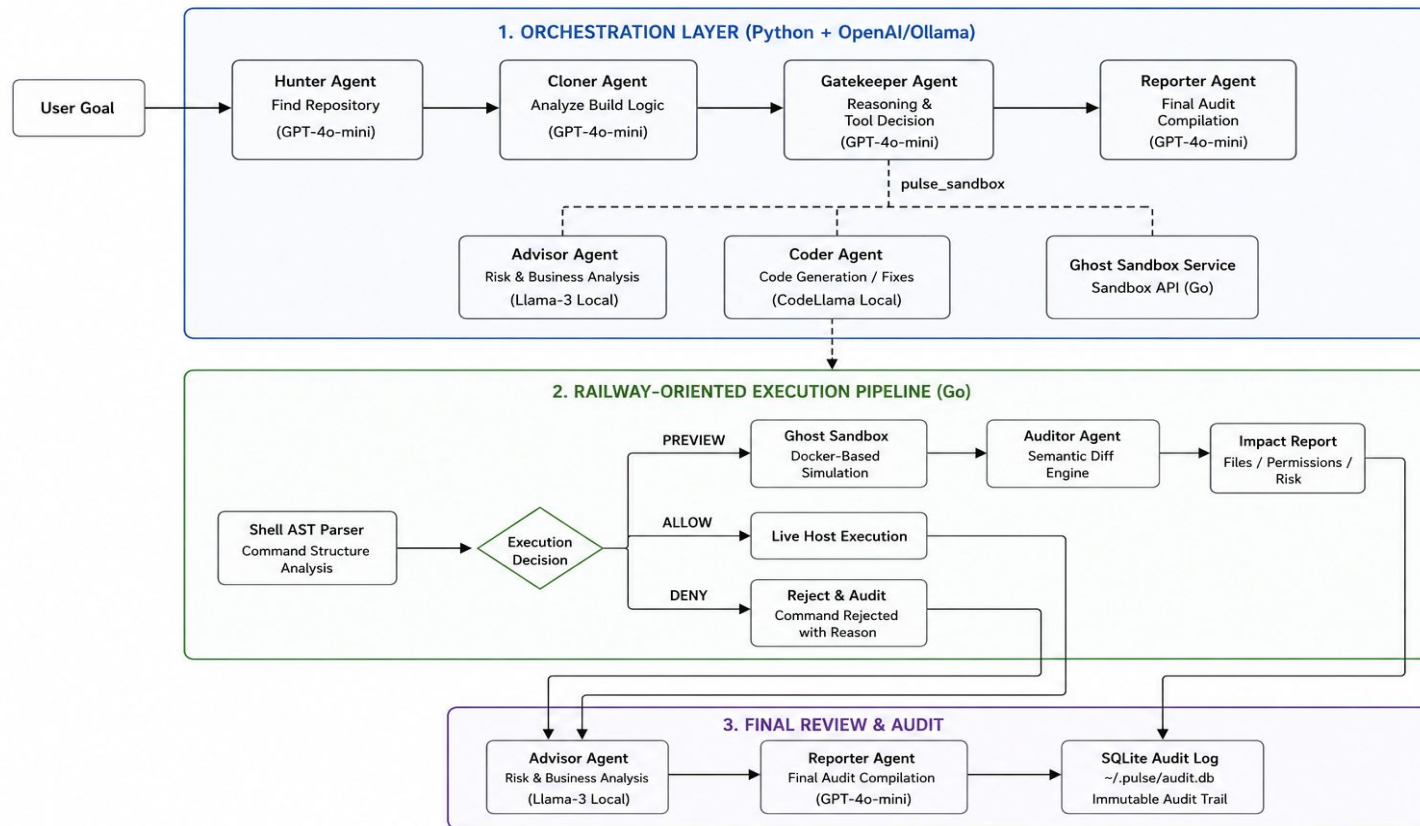
```
Welcome to Pulse (Dike) - Type 'exit' or 'quit' to leave.
pulse> pulse> rm -rf /
--- [Gatekeeper Agent]: Command is safe.
bash: pulse: command not found
Command error: exit status 127
--- [Auditor Agent]: Audit log recorded.
pulse>
pulse> pulse> rm -rf /
--- [Gatekeeper Agent]: Command is safe.
bash: pulse: command not found
Command error: exit status 127
--- [Auditor Agent]: Audit log recorded.
pulse> ^C
Aborted
(base)
```

Tech & Architecture

Component	Technology	Specific Usage	Rationale for Selection
Orchestrator	Python 3.11	Multi-agent ReAct loops and model routing logic.	High library support for AI/LLM SDKs and rapid prototyping of agentic workflows.
Backend Engine	Go (Golang)	Railway-Oriented Pipeline and low-level system calls.	Superior concurrency model and type safety for critical security-first infrastructure.
Cloud LLM	GPT-4o-mini	Primary agent orchestration and fast reasoning tasks.	High-speed inference with near-zero token cost for general coordination tasks.
Local LLM	Llama-3 (8B)	Deep risk advisory and business impact analysis.	Ensures user data privacy by performing deep-reasoning on-device via Ollama .
Code Intelligence	CodeLLaMA	Infrastructure-as-Code (IaC) and shell script review.	Optimized for code understanding; runs locally to prevent proprietary script leakage.
Isolation Layer	Docker Engine	Ghost Sandbox execution for command simulation.	Industry-standard containerization providing a lightweight, reproducible environment.
Shell Parser	mvdan.cc/sh	Shell AST (Abstract Syntax Tree) parsing.	Prevents shell injection and allows for intent analysis before the command is even run.
Audit Storage	SQLite	Immutable event logging and session state history.	Zero-configuration, local-first database that ensures audit trails are portable and resilient.
Sandbox OS	Alpine Linux	Minimal footprint OS inside the Ghost Sandbox.	Extremely low overhead (5MB image) allowing for sub-second sandbox cold-starts.
State Sync	Python-Dotenv	Secure environment variable management.	Standardizes configuration across development and production environments.



Tech & Architecture



Impact & Use cases

Real Line Scenarios

The "Fat-Finger" Deployment Error A developer accidentally types `rm -rf /` instead of a local directory. Pulse intercepts and simulates the command in the Ghost Sandbox, revealing that 15,000 system files would be deleted. This prompts an immediate block, saving the server from a crash.

The Disgruntled Ex-Employee An administrator tries to run `db.dropDatabase()` on a production instance using cached credentials. The Gatekeeper Agent identifies the high-risk intent and routes it to simulation, where the Auditor flags the data loss event and prevents the malicious wipe.

The Rogue Autonomous AI Agent An AI agent tasked with fixing a Dockerfile accidentally attempts to reformat the host's primary disk partition. Pulse acts as a "Deterministic Brake," sandboxing the script and identifying the destructive change before it can impact the live environment.

The Malicious Dependency Hook A developer runs `npm install` on a package containing a hidden script designed to exfiltrate environment variables. The Shell AST Parser detects the unauthorized network egress and file access, revealing the malicious "Blast Radius" before the host is compromised.

Scale Potential

Multi-Server Cluster Orchestration Pulse can scale from a single-node protector to a distributed "Security Mesh" that synchronizes audit logs and safety policies across thousands of cloud instances simultaneously.

Enterprise Policy Engine The system can expand into a global marketplace where organizations download and deploy specialized "Agentic Guardrails" for specific stacks like Kubernetes, AWS, or Azure.

Edge-Device Security By optimizing local Ollama integration, Pulse can be deployed on IoT devices and edge gateways, providing autonomous, on-device security for the industrial internet backbone.

Differentiation

Why Pulse STANDS OUT

Predictive Prevention Over Reactive Recovery Unlike SIEM tools or backup software that act after a disaster, Pulse intercepts and simulates the "Blast Radius" in real-time, preventing outages before they ever reach the host.

High-Fidelity Evidence (The Ghost State) Pulse replaces "AI Hallucinations" with "Simulated Reality." It provides empirical filesystem diffs from an isolated Docker replica, showing exactly what a command will do rather than guessing.

Frictionless Velocity By automating security verification for thousands of CLI commands per second, Pulse eliminates traditional security bottlenecks, allowing developers to move at peak speed with a guaranteed safety net.

Competitive MOAT

The Simulation-Logic Integration (The Ghost Moat)

Pulse has built a proprietary bridge between high-level Agentic Reasoning and low-level Container State Simulation. While competitors only "read" commands, Pulse "rehearses" them. This integration of a Go-based Railway Pipeline with a Docker-cloned filesystem creates a technical barrier that simple LLM-wrappers cannot replicate.

Hybrid Privacy-First Intelligence (The Inference Moat)

Pulse maintains a strategic advantage through its Multi-Model Routing Architecture. By offloading sensitive infrastructure analysis to local Ollama models while using Cloud LLMs for orchestration, we create a data-sovereignty moat. We provide enterprise-grade safety at near-zero token cost, a sustainable ratio cloud-only competitors cannot match.

Demo Link

- Demo Link :
<https://github.com/aryawadhwa/Dike>
- Document submitted